Unit 5

Environmental Policies & Practices : Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture, Environment Laws: Environment Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act, International agreements: Montreal and Kyoto protocols and Convention on Biological Diversity (CBD), Nature reserves, tribal populations and rights, and human wildlife conflicts in Indian context

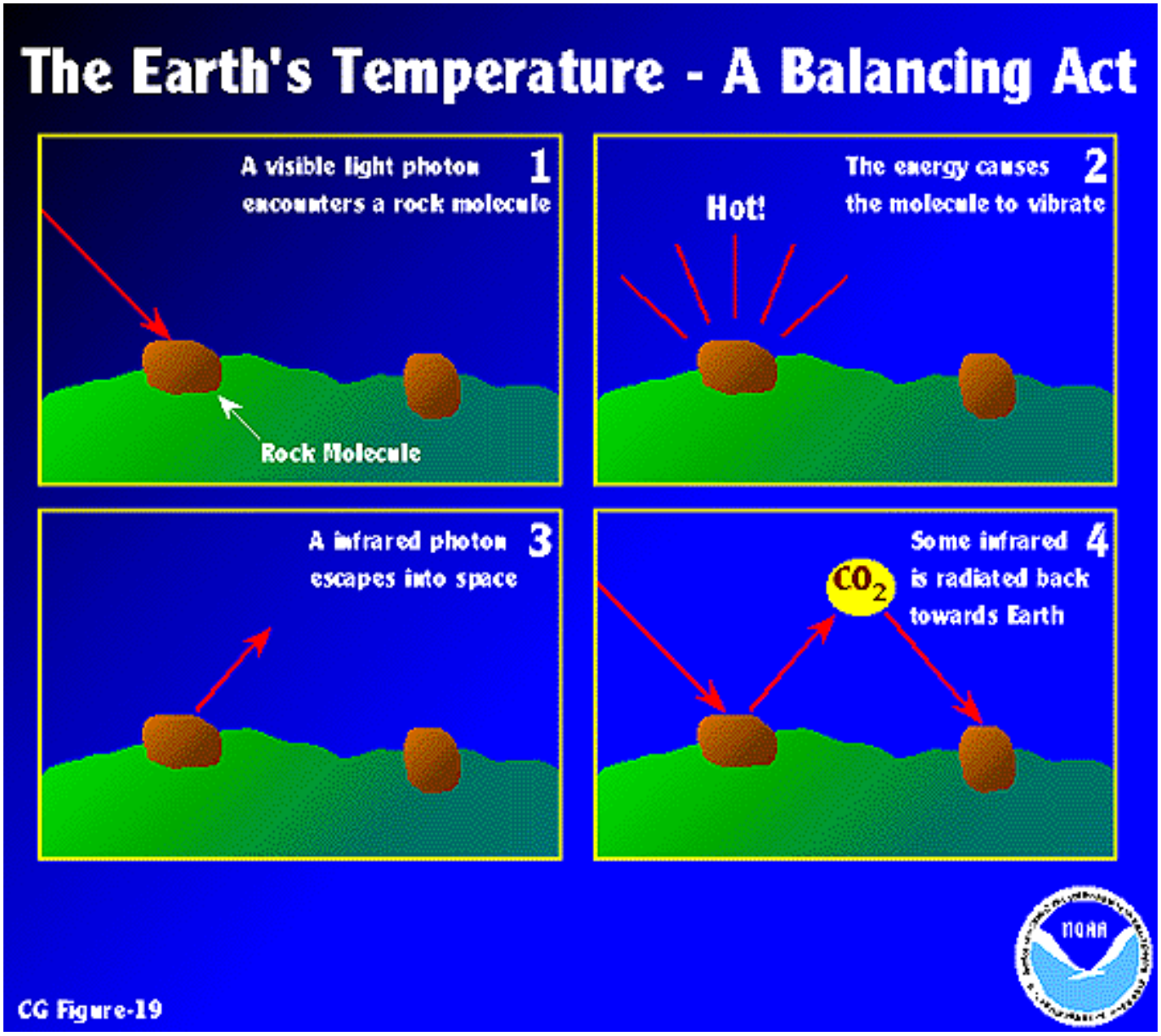
Environmental Laws and Policies

Dr. Dipakshi Sharma

Greenhouse Effect and Global Warming

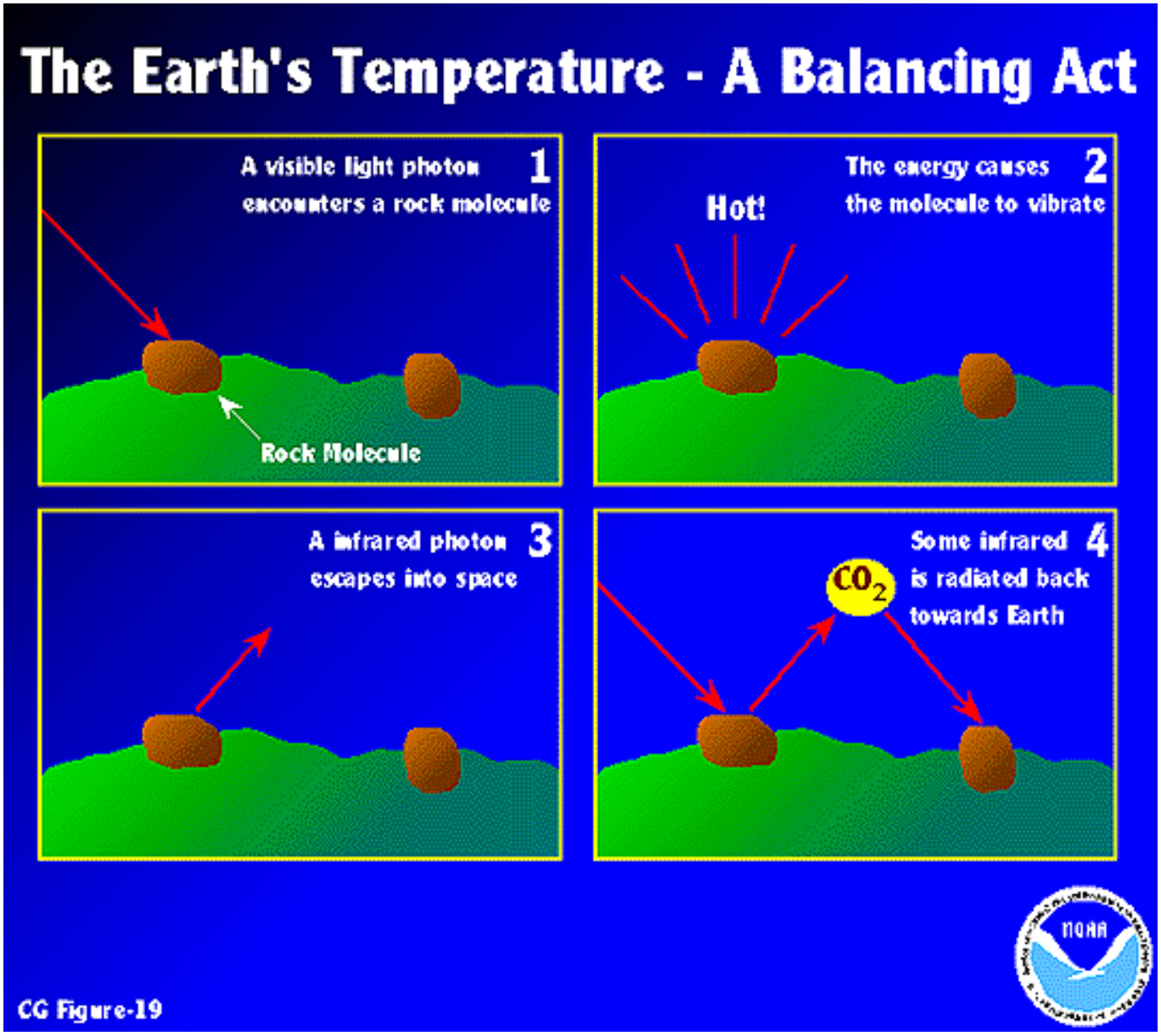
The term *greenhouse effect* is used to indicate a heat-trapping process caused by gases such as carbon dioxide, and water vapour which are transparent to incoming solar radiations but re-emit the infrared radiations from Earth's surface.

Global warming is a long-term rise in the average temperature of Earth as a whole as a result of greenhouse effect.



1. Shorter, high Energy wavelengths Hit the earths Surface

2. Incoming energy Is converted to heat



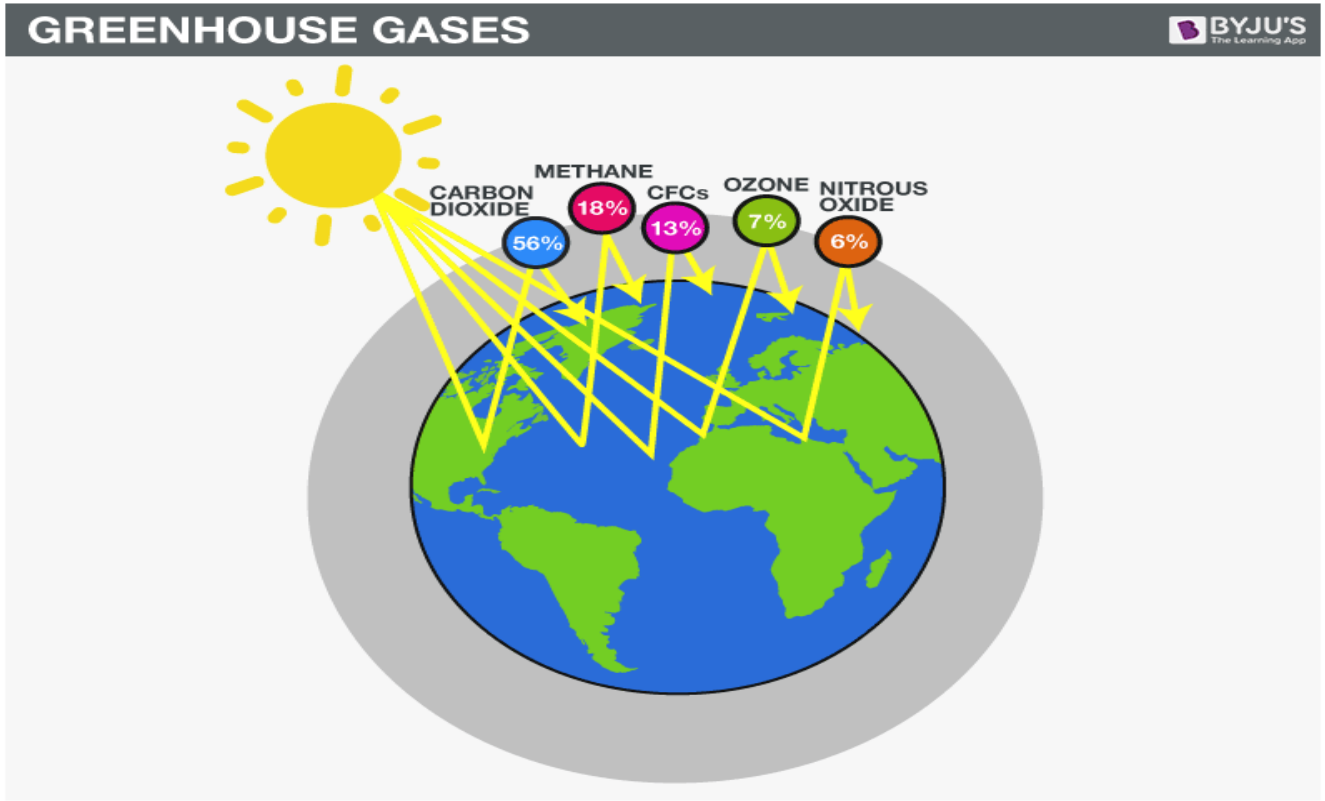
3. Longer, infrared Wavelengths hit Greenhouse gas Molecules in the atmosphere

4. Greenhouse gas Molecules in the Atmosphere emit Infrared radiation Back towards earth

Greenhouses Gases

Major Greenhouse Gases

- Carbon dioxide
- Methane
- Nitrous oxide
- Chlorofluoro carbons

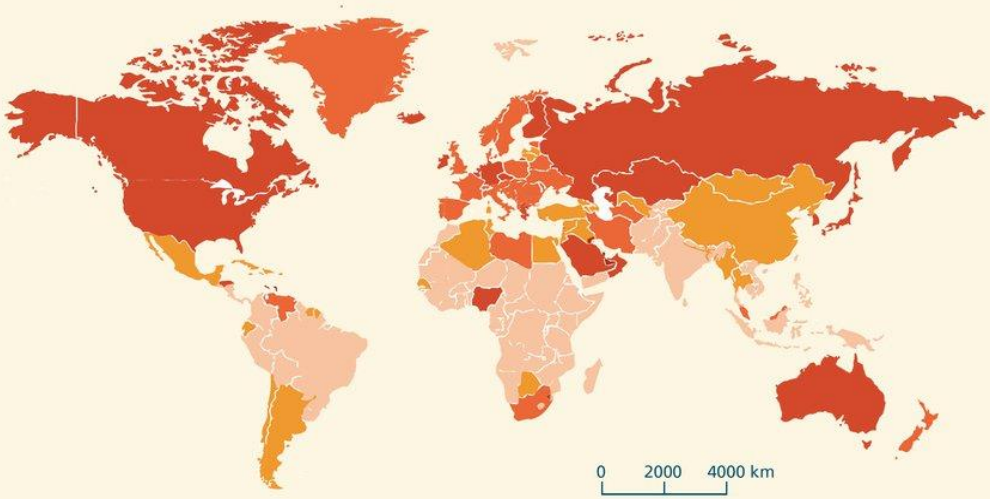


Sources of Greenhouse Gases

- Fossil-fuel burning
- Industrial processes
- Deforestation
- Livestock
- Biomass burning
- Coal mining

World Greenhouse Gas Emissions

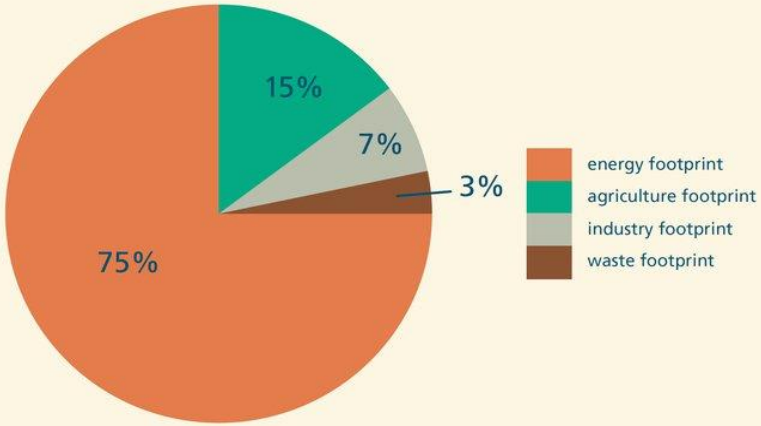
CO_2 emissions per capita in 2007



Unit tonnes of CO_2

0.02 – 2.00 | 2.01 – 5.00 | 5.01 – 10.00 | 10.01 – 30.00 | 30.01 – 60.00

Breakdown of Australia's carbon footprint



Highest carbon footprint per capita

Australia | USA | Canada | Korea

Carbon footprint of different foods

18kg CO_2	4.6kg CO_2	0.15kg CO_2	0.3kg CO_2	0.55kg CO_2	0.38kg CO_2
1kg of beef	1kg of chicken	1 egg	1 glass of milk	1 kg of apples	1 kg of bread

Museum2you

NSW GOVERNMENT

NSW Environmental TRUST

Australian museum
nature culture discover

Global Warming Potential

It is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO_2)

- The time period usually used for GWPs is 100 years
- The 20-year GWP is sometimes used as an alternative to the 100-year GWP. This 20-year GWP prioritizes gases with shorter lifetimes

GHG	GWP for 100 years
CO_2	1
CH_4	23
N_2O	296
HFC - 23	12 000
HFC – 134a	1 300
SF_6	22 200

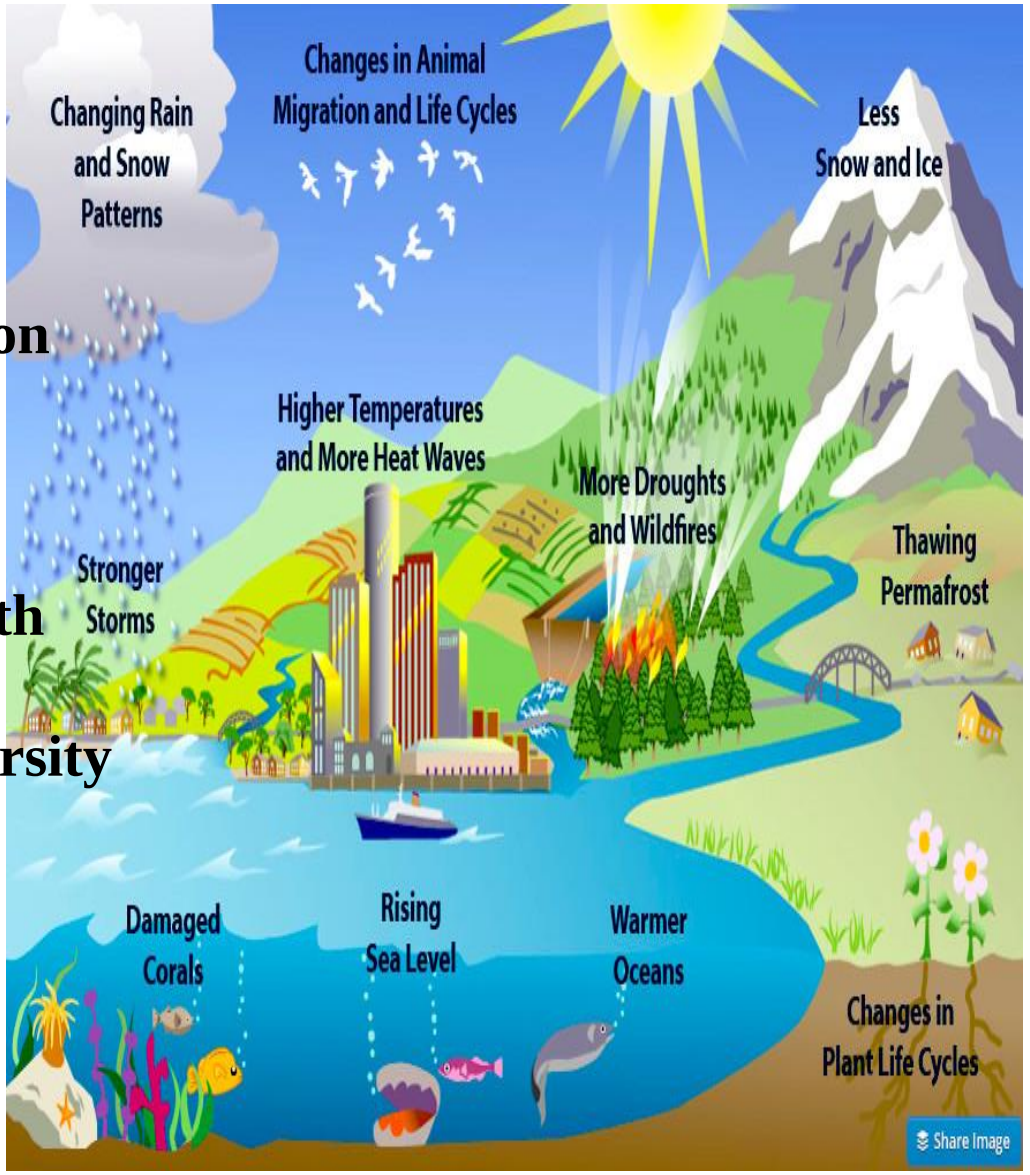
Source: *IPCC Third Assessment Report* (2001).

Effects of CO_2

- Air and the Earth's surface may grow warmer.
- The stratosphere may become cooler.
- Temperate and polar regions may become warmer leading to the reduction in the ice cover of the earth.
- Rainfall may be higher than what it is present in the temperate regions.
- The greater amount of evaporation due to excess warmth .

Environmental Effects of Global warming

- Climate change
- Rise in sea level
- Reduced agricultural production
- Storms
- Adverse effects on human health
- Loss of ecosystems and biodiversity



Control measures

- Reduction in the use of fossil fuels
- Shifting to the renewable energy sources that do not emit greenhouse gases
- Increasing the use of energy efficient and cleaner production technologies and practices
- Reducing deforestation, adopting better forest management practices, and undertaking aforestation to sequester carbon

Acid Rain

Acid rain refers to a condition in which natural precipitation becomes acidic after reacting chemically with pollutants in the air.

Causes of Acid Rain

Burning of the fuel as well as the use of nuclear weapons leading to the evolution of large amounts of sulphur dioxide (SO_2) and nitrogen dioxide (NO_2), which get converted to sulphurous acid and nitric acid, respectively.

Effects of Acid Rain

- Reduction in population of aquatic biota
- Damage to terrestrial ecosystems
- Corrosion of buildings
- Effect on human beings

Chemical Processes Involved In acid rain

Formation Of Sulphuric Acid

$$S + O_2 \longrightarrow SO_2$$

$$SO_2 + 1/2O_2 + H_2O \longrightarrow H_2SO_4$$

Reaction Involving Formation Of Nitric Acid

$$NO + O_3 \longrightarrow NO_2 + O_2$$

$$NO_2 + O_3 \longrightarrow NO_3 + O_2$$

$$NO_3 + NO_2 \longrightarrow N_2O_5$$

$$N_2O_5 + H_2O \longrightarrow 2HNO_3$$

Ozone Layer and Its Depletion

- The decrease in the concentration of ozone (O_3) in stratosphere is known as ozone layer depletion.
- It is first observed in Antarctic continent and known as ozone hole spreading over approximately seven million square kilometer.
- It is measured in Dobson Units (DU)
- It is 3 mm in thickness or about 300 Dobson Units.
- 16th of September is observed as the World Ozone Day
- Chlorofluorocarbons (CFCs) as aerosol propellants, refrigeration, fire suppression and foam insulation etc.

Mechanism of Ozone Layer Depletion

a) $$CFC \xrightarrow{UV} Cl^{\bullet}$$

b) $$O_2 \xrightarrow{UV\ C} O^{\bullet} + O^{\bullet}$$

$$O_2 + O^{\bullet} \longrightarrow O_3$$

$$O_3 \xrightarrow{UV\ B} O_2 + O^{\bullet}$$

$$Cl^{\bullet} + O^{\bullet} \longrightarrow ClO$$

c) $$ClO + ClO \longrightarrow ClOOCl\ (\text{dimer})$$

d) $$ClOOCl \longrightarrow Cl_2 + O_2$$

e) $$Cl_2 \longrightarrow Cl^{\bullet} + a$$

Environmental Effects of O_3 Layer Depletion

- Damage to land plants
- Decrease in population of aquatic community
- Increase in the rate of skin diseases and cancers in human beings
- Disturbance in climate patterns
- Corrosion of materials

Control Measures

- Evolving substitutes for CFCs
- Development of hydrochloroflourocarbons and hydrofluorocarbons

Montreal Protocol- Ozone layer

To phase down the consumption and production of the different Ozone depleting Substances (ODS) in a step-wise manner, with different timetables for developed and developing countries

- Adopted on 16 September 1987
- The Parties to the Protocol have phased out 98% of ODS globally compared to 1990 levels
- From 1990 to 2010, the treaty's control measures are estimated to have reduced greenhouse gas emissions by the equivalent of 135 gigatons of CO2, the equivalent of 11 gigatons a year.

- Overall HFC emissions are growing at a rate of 8% per year and annual emissions are projected to rise to 7-19% of global CO_2 emissions by 2050.
- HFC emissions, therefore, challenges efforts to keep global temperature rise at or below 2°C this century a target agreed at the Paris climate conference.
- Hydrochlorofluorocarbons (HCFCs) are gases used worldwide in refrigeration, air-conditioning and foam applications etc.

MONTREAL PROTOCOL CONTRIBUTES TO THE SUSTAINABLE DEVELOPMENT GOALS

Kyoto Protocol-Global Warming and Climate Change

- The Kyoto Protocol to the United Nations Framework Convention on climate change is an amendment to the international treaty on climate change
- **Objective:** "Stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climatic system"
- The Kyoto Protocol was adopted on **11 December 1997**
- It entered into force on **16 February 2005** and Currently, there are **192 Parties to the Kyoto Protocol.**
- It promoted the establishment of **flexible market mechanisms,** which are based on the trade of emissions permits

1. **International Emissions Trading**
2. **Clean Development Mechanism (CDM): Trading of Carbon** i.e. **Carbon credit/Carbon point**

1 Carbon Credit = 1 ton of CO_2 gas emitted

Convention on Biological Diversity(CBD)

Goals of the Convention:

1. Conservation of Biological Diversity

2. Sustainable use of the components of the Biodiversity

3. Fair and equitable sharing of benefits arising from the genetic resources

- The Secretariat of the Convention on Biological Diversity (SCBD) is based in Montreal, Canada.
- It is ratified by 196 nations.
- It covers biodiversity at all levels: **Ecosystems, Species and Genetic resources**
- The treaty meets every two years to review progress, set priorities and commit to work plans.

Convention on Biological Diversity(CBD)

- The Conference of Parties (COP) is the governing body of the convention. It consists of the governments that have ratified the treaty.
- The Convention was opened for signature on 5 June 1992 at the United Nations Conference on Environment and Development (the Rio "Earth Summit")
- The Convention entered into force on 29 December 1993,
- 196 countries are a party to the CBD
- India is also a party to the Convention. India ratified it in 1994.

The Cartagena Protocol on Biosafety

The **safe handling, transport and use of living modified organisms (LMOs) resulting from modern biotechnology that may have adverse effects on biological diversity**, taking also into account risks to human health. It was adopted on **29 January 2000** and entered into force on **11 September 2003.**

The Nagoya – Kuala Lumpur Supplementary Protocol on Liability and Redress to the Cartagena Protocol on Biosafety

- **The conservation and sustainable use of biodiversity by providing international rules and procedures in the field of liability and redress relating to living modified organisms.**
- Adopted on **15 October 2010** at the fifth meeting of the Conference of the Parties serving as the meeting of the Cartagena Protocol on Biosafety, held in Nagoya, Japan, and entered into force on **5 March 2018**.

Environmental Acts

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History of Environmental Laws

India's original Constitution did not contain any provision for the protection of the natural environment. However, **the Fundamental Duties**, which were added by the **42nd Amendment to the Constitution**, prescribed the protection of the environment including **forests, lakes, rivers and wildlife as a duty of the citizens of the country**

- These developments were a result of the United Nations Conference on Human Environment, held in Stockholm in 1972.
- **The Wildlife Protection Act, 1972.**
- **The Water (Prevention and Control of Pollution) Act, 1974**
- **The Air (Prevention and Control of Pollution) Act 1981.**
- In 1986, **the EPA** was passed by the Parliament, in the wake of the Bhopal Gas Tragedy, which occurred on Dec 2, 1984

The Environment Protection Act (1986)

To protect and improve the environment and environmental conditions.

Provisions of Environment Protection Act

- It can coordinate and execute nationwide programmes and plans to further environmental protection.
- It can mandate environmental quality standards, particularly those concerning the emission or discharge of environmental pollutants.
- This law can impose restrictions on the location of industries.

The Environment Protection Act (1986)

- The law gives the government the **power of entry for examination, testing of equipment and other purposes and power to analyse the sample of air, water, soil or any other substance** from any place.
- The EPA explicitly bars **the discharge of environmental pollutants in excess of prescribed regulatory standards**.
- There is also in place a specific provision for **handling hazardous substances,** which is prohibited unless in compliance with regulatory requirements.
- The Act empowers any person, apart from authorised government officers, to file a complaint in a court regarding any contravention of the provisions of the Act.

Aims/Objectives of the Environment Protection Act

- Implementing the decisions made at the **United Nations Conference on Human Environment held in Stockholm.**
- **Creation of a government authority to regulate industry** that can issue direct orders including closure orders.
- Coordinating activities of different agencies that are operating under the existing laws.
- Enacting regular laws for the protection of the environment.
- **Imposing punishments and penalties** on those who endanger the environment, safety and health. For each failure or contravention, the punishment includes **a prison term of up to five years or a fine of up to Rs. 1 lakh, or both.** This can also be **extended for up to seven years in cases.**
- Engaging in the sustainable development of the environment.
- Attaining protection of the **right to life under Article 21 of** the Constitution.

Air (Prevention and Control of Pollution) Act- 1981

Objective- It provides the prevention , control and abatement of air pollution. Provisions of the Act has to be implemented by Central Pollution Control Board along with the State Pollution Control Board.

Functions-

1. Setting of the air quality standards, collecting data on air pollution, organizing training, awareness programmes, establishing laboratories
2. Specify air pollution control areas and set standard for vehicle emissions.
3. Penalties for violation of its provisions. Applies to all.

Water (Prevention and Control of Pollution) Act 1974

Objectives-

1. It prevents, controls and provides maintenance or restoration of wholesomeness of water.
2. Provisions of the Act has to be implemented by Central Pollution Control Board being a main agency.

Functions-

1. Establishes standards of the water quality and effluent
2. Polluting industries must seek the permission to discharge the waste into effluent bodies.

Wildlife Protection Act- 1972

Objectives-

Wildlife protection Act of 1972 defines wildlife to include any bird or animal and aquatic or land vegetation, form part of any habitat.

Function-

1. Under the Act, Central government work with state governments to regulate or prohibit the conversion of forest in agriculture or urban land.
2. Protection against natural hazards
3. Maintenance of water supply in water bodies present in Forest.
4. Protection of lines of communication and transportation

Forest Conservation Act of 1980

Objectives-

1. It provides for the protection and conservation of the forests.
2. Under the Act, a state government may regulate or prohibit in any forest the clearing of land for cultivation, pasturing of cattle, or clearing the vegetation for any of following purposes:

Function-

1. Protection against natural hazards
2. Maintenance of water supply in aquifers.
3. Protection of lines of communication and transportation
4. Preservation of public health.
5. Mandatory for land owner.

National Forest Policy 1988

Aim: to ensure environmental stability and maintenance of ecological balance including atmospheric equilibrium which are vital for sustenance of all life forms, human, animal and plant.

- Major achievements of National Forest Policy

Increase in the forest and tree cover.

- Involvement of local communities in the protection, conservation and management of forests through Joint Forest Management Programme.
- Meeting the requirement of fuel wood, fodder minor forest produce and small timber of the rural and tribal populations.
- Conservation of Biological Diversity and Genetic Resources of the country through ex-situ and in-situ conservation measures.
- Significant contribution in maintenance of environment and ecological stability in the country.

Objective of National Forest Policy

Maintenance of environmental stability through preservation and, where necessary, restoration of the ecological balance that has been adversely disturbed by serous depletion of the forests of the country.

- Conserving the natural heritage of the country by preserving the remaining natural forests with the vast variety of flora and fauna, which represent the remarkable biological diversity and genetic resources of the country.
- Checking soil erosion and denudation in the catchments areas of rivers, lakes, reservoirs in the “interest of soil and water conservation, for mitigating floods and droughts and for the retardation of siltation of reservoirs.
- Checking the extension of sand-dunes in the desert areas of Rajasthan and along the coastal tracts.
- Increasing substantially the forest/tree cover in the country through massive afforestation and social forestry programmes, especially on all denuded, degraded and unproductive lands.
- Meeting the requirements of fuel-wood, fodder, minor forest produce and small timber of the rural and tribal populations.
- Increasing the productivity of forests to meet essential national needs.
- Encouraging efficient utilisation of forest produce and maximising substitution of wood.
- Creating a massive people’s movement with the involvement of women, for achieving these objectives and to minimise pressure on existing forests.

Issues involved in enforcement of Environmental Laws

1. Illiteracy
2. Growing population
3. Ignorance
4. Economic reasons
5. Insufficiency of laws

Issues Involved in Enforcement of Environmental Legislation in India

1.**The precautionary principle:** This principle has evolved to deal with risks and uncertainties faced by environmental management. The principle implies that an ounce of prevention is better than cure it does not prevent problems but may reduce their occurrence and helps ensure contingency plans are made.

-But some environmental problems become impossible or costly to solve if there is delay, therefore waiting for research and legal proof is not costless.

2. The polluter-pays principle: The polluter pays for the damaged caused by a development. This principle also implies that a polluter pays for monitoring and policing.

-A problem with this approach is that fines may bankrupt small businesses, yet be low enough for a large company to write them off as an occasional overhead, which does little for pollution control.

3. Freedom of information: Environmental planning and management is hindered if the public, NGOs or even official bodies are unable to get information. Many countries have now begun to release more information, the USA has a Freedom of Information Act, and the European Union is moving in this direction.

-But still many governors and multinational corporations fear that industrial secrets will leak to competitors if there is too much disclosure, and there are situations where authorities declare strategic needs and suspend disclosure.

Human - wildlife conflict in India

Human wild life conflict in India

Tribal population in India and Their Rights

Tribe

A Tribe is a group of distinct people, dependent on their land for their livelihood, who are largely self-sufficient, and not integrated into the National society. They are considered to be existing in the native area’s as a social group existing before the development of Nation States, or outside them.

India's forests are home to crore’s of people, including many Scheduled Tribes, who live in or near the forest areas of the country. Nearly 250 million people live in and around forests in India, of which the estimated indigenous Adivasi or tribal population stands at about 100 million.

Tribes Are also called Scheduled- Why?

The reason why they are called **SCHEDULE** is because they are included in **one of the Schedule of the Constitution of India.** Our Constitution has **12 Schedules**. The Scheduled Tribes people are the one who are previously untouchables. According to Hindu Mythology, this is the fifth category in the VARNA system. The Scheduled Tribes people are generally called as ADIVASI.

Problems Related to Scheduled Tribes

1) Land Rights: Under the 2006 Forest Rights Act, providing land rights to forest-dwelling communities of the country, only 15 lakh complainants out of the 39,56,262 cases filed; were given the legal recognition to their property.

2) The tribes have their own social problems. They are traditional and bound by customs.

3) Illitracy: Due to lack of education and development they are superstitious and believe in outmoded and meaningless practices which can be harmful at times.

4) Others: Child marriage, infanticide, homicide, animal sacrifice, exchange of wives, black magic and other detrimental practices are still prevalent among them. They believe in supernatural powers and have a keen desire to maintain these practices. They don't want to change their significant tribal character and hence it's said that "tribes are the tribesmen first, the tribesmen last, and the tribesmen all the time. "

Constitutional Rights to Tribal People

The Constitution of India has provided special provisions to the tribal people to safeguard their interests:

1) Article 15 of the Indian Constitution states that the state shall not discriminate any citizen on grounds of religion, race, caste, sex, place of birth or any of them. This explains that every citizen of India is provided equal rights and opportunities without any discrimination.
2) Government of India has made reservation for the tribes in employment under Article 16(4) of the Constitution of India.
3) The Government of India has reserved seats in The House of People (Lok Sabha) and The State Legislative Assemblies under Article 330 and 332 of The Constitution of India.
4) Article 19(5) of the Constitution of India guarantees the tribal people right to own property and enjoy it in any part of the country.
5) Article 338 of The Constitution of India grants the right to appoint a Commissioner to look after welfare activities of tribes.
6) Article 46 of the Constitution of India states that, The State shall promote with special care the educational and economic interests of the weaker sections of the people and in particular, the Scheduled Castes and the Scheduled Tribes, and shall protect them from social injustice and all forms of exploitation.
7) Under Article 275 of the Constitution of India the Centre Government is required to give grants-in-aid to the State Government for approved Tribal Welfare Schemes.

Importance of tribal communities

1) As they are living in close association with nature they carry inherent knowledge of plants and animals that help us to stratergise protection measures.

2) They count to diversity of India, culture, cuisine and medication (Ethnomedication).

3) They are providing protection to number of plants and animals species from hunter and traders.

4) Number of tribal people are becoming role models around the globe and set an extraordinary examples in protecting environment and biodiversity.

Ex. The Kondh community who marched into the streets of Bhubaneshwar to prevent their nature and their living area from potential environmental violatiors.

Simple steps to reduce the emission of GHC

Reduce, Reuse, Recycle
Use Less Heat and Air Conditioning
Replace Your Light Bulbs (Incandescent bulb to CFL/LED)
Drive Less and Drive Smart
Buy Energy-Efficient Products
Use Less Hot Water
Energy conservation
Plant a Tree (Afforestation)
Encourage Others to Conserve
Development of Clean Technologies
Reduction in Carbon footprints